INCH-POUND

MIL-PRF-1/894F
10 April 2007
SUPERSEDING
MIL-E-1/894E
21 November 1985

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 884

INACTIVE FOR NEW DESIGN
AFTER 30 APRIL 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification and MIL-PRF-1.

DESCRIPTION: Triode.

Mounting position: Any.

Weight: 1.5 ounces (42.5 grams) nominal.

Envelope: ST-12 (EIA).

Base: B6-3 (EIA).

Height: 4.125 inches (104.78 mm) maximum.

Diameter: 1.563 inches (39.70 mm) maximum.

Pin connections:

Pin no.:	1	2	3	5	7	8
Element:	nc	h	a	g	h	k

ABSOLUTE RATINGS:

Parameter:	Ef	epx or epy	Peak voltage between any two electrodes	ib	Ib	Rg	tk	Ehk	TA
Unit:	V	v	v	mA	mA dc (Note 1)	MegΩ (Note 2)	s	V dc (Note 3)	°C
Oscillator:									
Maximum:	6.93	300	350	300	2	0.5	30	-100	+85
Minimum:	5.67	---	---	---	---	---	---	---	-60
Rectifier:									
Maximum:	6.93	300	350	300	75	0.5	30	-100	+85
Minimum:	5.67	---	---	---	---	---	---	---	-60
Test conditions:	6.3	---	---		---	---	30	---	---

GENERAL:

Qualification - Not required.

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TABLE I. Requirements or tests.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 1	4/					
3241	Heater current			If	540	660	mA
3201	Critical grid voltage for conduction (1)	5/	Eb = 110 V dc; Rp = 1,700 ohms; Rg = 0.1 MegΩ	Ec(1)	-9	-16	V dc
3201	Critical grid voltage for conduction (2)		Eb = 250 V dc; Rp = 5,600 ohms; Rg = 0.1 MegΩ	Ec(2)	-21	-31	V dc

1231	Pulse emission		epp = eggl = 180 ± 9.0 v;
			prr = 100 ± 5;
			tk = 5 (min);
			Ra = 10 ohms;
			Rp = 20 ohms;
			calibrating resistor = 10 ohms;
			Zm = 7.5;
			t = 3 (max)
				etd	---	69	v

	Conformance inspection, part 2						
1031	Low frequency vibration		No voltages applied	---	---	---	---
1336	Heater-cathode leakage			Ihk	0	20	µA dc
	Conformance inspection, part 3						
---	Life test		Group B (see figure 1); t = 500 hours	---	---	---	---
---	Life-test end points:						
3201	Critical grid voltage for conduction (1)		Eb = 110 Vdc; Rp = 1,700 ohms; Rg = 0.1 MegΩ	Ec(1)	-7.0	-18.0	V dc
3201	Critical grid voltage for conduction (2)		Eb = 250 Vdc; Rp = 5,600 ohms; Rg = 0.1 MegΩ	Ec(2)	-19.0	-33.0	V dc

1231	Pulse emission		epp = eggl = 180 ± 9.0 v;
			prr = 100 ± 5;
			tk = 5 (min);
			Ra = 10 ohms;
			Rp = 20 ohms;
			calibrating resistor = 10 ohms;
			Zm = 7.5;
			t = 3 (max)
				etd	---	110	v

See footnotes at top of next page.

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TABLE I. Requirements or tests - Continued.

1/ Below 200 Hz, Ib = 3 mA dc (maximum).

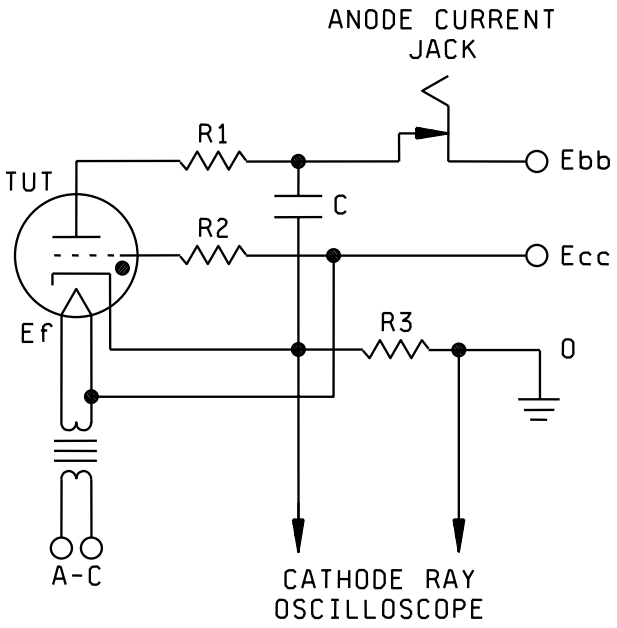
2/ Grid resistance shall be not less than 1,000 ohms per volt (maximum instantaneous) applied to grid.

3/ Peak voltage between any electrode and heater shall not exceed 350 volts.

4/ The acceptance level (see note 6) for all tests listed in conformance inspection, part 1, shall be 0.65.

5/ This test shall be the first test performed at the conclusion of the holding period.

6/ This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.



Ebb = 300 V dc
Ecc = 150 V dc
R1 = 500 OHMS
R2 = 30,000 OHMS
R3 = 330,000 OHMS
C = 0.005 µF

PEAK VOLTAGE - 136 VOLTS
PEAK CURRENT - 250 mA
FREQUENCY - 1,000 Hz
AVERAGE CURRENT - 0.7 mA dc

FIGURE 1. Relaxation oscillator test circuit.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-2007-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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